An Algebraic Approach to the Hydrogen Atom

Introduction

Historically, the hydrogen spectrum was one of the key empirical puzzles of late 19th and early 20th century physics. Experiments showed that hydrogen emits light at discrete wavelengths, summarized by formulas like Balmer's (1885) and later generalized by Rydberg. Niels Bohr (1913) provided the first theoretical explanation using quantized electron orbits, giving the energy levels

$$E_n = -\frac{13.6 \text{ eV}}{n^2}.$$

Bohr's model was a major step forward, but from a theoretical standpoint it was unsatisfactory. The biggest issue was that it relied on ad hoc postulates rather than a coherent underlying theory. This made it difficult to systematically refine the model and explain certain phenomena such as fine structure, Zeeman splitting, and the Stark effect.

In 1926, Wolfgang Pauli used Heisenberg's matrix mechanics (dubbed the new quantum theory) to solve the hydrogen atom before Schrödinger's wave mechanics had fully formed. Pauli used symmetry arguments and what we now recognize as hidden conservation laws (related to the Runge–Lenz vector) to derive the hydrogen energy levels algebraically, without solving a differential equation. The methodology is very similar to the strategy we used to solve for the eigenstates of angular momentum and the Hamiltonian of a quantum harmonic oscillator! His work helped solidify the internal consistency of early quantum mechanics and deepened our understanding of why the hydrogen spectrum has the structure it does, beyond simply reproducing the formula.

We begin with a review of the classical Kepler problem for a particle in a $1/r$ potential, emphasizing the role of conserved quantities. After introducing angular momentum, we construct the Runge–Lenz vector and show how its conservation leads to an algebraic characterization of the orbits, revealing a hidden symmetry of the system. We then turn to the quantum mechanical treatment of the hydrogen atom in the electrostatic (Coulomb) model. Rather than solving the Schrödinger equation directly, we follow an algebraic approach in the spirit of Pauli's work. We introduce the operator analog of the Runge–Lenz vector, and analyze its commutation relations. Finally, we use the algebraic structure to derive the hydrogen energy spectrum.

Classical Kepler Problem

We consider an attractive central force given by the inverse-square law as

$$\mathbf{F} = -\nabla V = -\frac{\gamma}{r^2}\hat{\mathbf{r}}.$$

Examples include Newtonian gravity ($\gamma = GMm$) and Coulomb's law for the electrostatic force between the proton and the electron of a neutral hydrogen atom ($\gamma = k_e e^2$). We consider the limit of a two-body problem where one mass is very light compared to its interaction partner. For example, the mass of the proton is roughly 1836 times that of the electron, so we will consider the proton to be effectively stationary. In the small m limit, the classical system has 6 degrees of freedom (3 for position and 3 for momentum). We search for *constants of motion*, meaning quantities depending on position and momentum that are guaranteed to be conserved by the laws of motion. Each constant of motion eliminates a degree of freedom, i.e., each constrains the allowed motions.

Total Energy

The total energy of a particle with mass m in a Kepler potential is

$$E = \frac{1}{2}m\mathbf{v}^2 - \frac{\gamma}{r}$$

Using Newton's second law, we can show that the total energy of the particle is conserved:

$$\dot{E} = m\mathbf{v}\cdot\dot{\mathbf{v}} + \frac{\gamma}{r^2}\dot{r} = \mathbf{v}\cdot\mathbf{F} + \frac{\gamma}{r^2}\dot{r} = -\mathbf{v}\cdot\frac{\gamma\hat{\mathbf{r}}}{r^2} + \frac{\gamma}{r^2}\dot{r} = \frac{\gamma}{r^2}(\dot{r} - \mathbf{v}\cdot\hat{\mathbf{r}})$$

In the last term, since $\dot{r} = \hat{\mathbf{r}}\cdot\mathbf{v}$, it follows that $\dot{E} = 0$. Conservation of energy constrains the relationship between v and r. For instance, as r increases, v must decrease.

Angular Momentum

The particle's angular momentum is defined as

$$\mathbf{L} = \mathbf{r}\times\mathbf{p} = m\mathbf{r}\times\mathbf{v}$$

and it is conserved for any central force. Since $\mathbf{r}$ is always parallel to $\mathbf{F}$,

$$\dot{\mathbf{L}} = \dot{\mathbf{r}}\times\mathbf{p} + \mathbf{r}\times\dot{\mathbf{p}} = \mathbf{r}\times\mathbf{F} = 0$$

So far, we have $1 + 3 = 4$ constants of motion; one is energy, and the other three are the components of angular momentum. By conservation of $\mathbf{L}$, we can demonstrate the orbit remains in a plane using cyclical symmetry of the scalar triple product:

$$\mathbf{r} \cdot \mathbf{L} = \mathbf{r} \cdot (\mathbf{r} \times \mathbf{p}) = \mathbf{p} \cdot (\mathbf{r} \times \mathbf{r}) = 0$$

$$\mathbf{p} \cdot \mathbf{L} = \mathbf{p} \cdot (\mathbf{r} \times \mathbf{p}) = \mathbf{r} \cdot (\mathbf{p} \times \mathbf{p}) = 0$$

The position and momentum are both perpendicular to the angular momentum, so they lie in the orbital plane defined by the angular momentum.

Runge–Lenz Vector

In classical mechanics, the Runge–Lenz vector is defined as

$$\mathbf{A} = \frac{1}{m\gamma} (\mathbf{p} \times \mathbf{L}) - \hat{\mathbf{r}}$$

While it's not as immediately obvious as for energy or angular momentum, it is indeed the case that the Runge–Lenz vector is a conserved quantity of the motion:

$$\begin{aligned}
\dot{\mathbf{A}} &= \frac{1}{m\gamma} \left[\frac{d\mathbf{p}}{dt} \times \mathbf{L} + \mathbf{p} \times \frac{d\mathbf{L}}{dt} \right] - \frac{d\hat{\mathbf{r}}}{dt} \\
&= \frac{1}{m\gamma} \left(-\frac{\gamma}{r^2} \hat{\mathbf{r}} \times (\mathbf{r} \times \mathbf{p}) \right) - \frac{d}{dt} \left(\frac{\mathbf{r}}{r} \right) \\
&= -\frac{1}{r} \hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \mathbf{v}) - \frac{\mathbf{v}}{r} + \mathbf{r} \frac{\dot{r}}{r^2} \\
&= -\frac{1}{r} \left[\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{v}) - \mathbf{v}(\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}) \right] - \frac{\mathbf{v}}{r} + \mathbf{r} \frac{\hat{\mathbf{r}} \cdot \mathbf{v}}{r^2} \\
&= 0
\end{aligned}$$

Having demonstrated that $\mathbf{A}$ is conserved, it is natural to wonder if and how it constrains the motion in a way that sets it apart from energy or angular momentum. The magnitude of $\mathbf{A}$ is already determined by the energy and angular momentum:

$$A^2 = 1 + \frac{2L^2}{m\gamma^2} E.$$

It must be the direction of $\mathbf{A}$ that somehow constrains the allowed motions of the particle. The Runge–Lenz vector lies in the same plane as $\mathbf{r}$ and $\mathbf{p}$ since $\mathbf{L} \cdot \mathbf{A} = 0$. Let's define the angle θ to be the angle between $\mathbf{r}$ and $\mathbf{A}$ in the orbital plane. In this way, $\mathbf{A}$ serves to define a reference line for the orbit. The projection of $\mathbf{r}$ onto $\mathbf{A}$ is given by

$$\mathbf{r} \cdot \mathbf{A} = \frac{L^2}{m\gamma r} - 1$$

and rearranging for the radial coordinate, we find

$$r = \frac{L^2/m\gamma}{1 + A\cos\theta}$$

This is the equation of the orbit! We can now identify A as the orbital eccentricity. The last remaining degree of freedom is encoded in the initial condition as the initial angular position of the particle along its otherwise fully specified orbit.

Quantum Runge–Lenz Operator

In the classical Kepler problem, the Runge–Lenz vector was a key that unlocked solutions in an elegant way. Define the Runge–Lenz operator (an operator-valued 3-vector) as

$$\mathbf{A} = \frac{1}{2m\gamma}\left(\mathbf{p} \times \mathbf{L} - \mathbf{L} \times \mathbf{p}\right) - \frac{\mathbf{r}}{r}.$$

Here are some key properties of $\mathbf{A}$, stated without proof for now:

$$A_i^\dagger = A_i \tag{1}$$

$$[H, A_i] = 0 \tag{2}$$

$$[L_i, A_j] = i\hbar\epsilon_{ijk}A_k \tag{3}$$

$$[A_i, A_j] = -\frac{2i\hbar H}{m}\,\epsilon_{ijk}L_k \tag{4}$$

$$\mathbf{A}^2 = 1 + \frac{2H}{m\gamma^2}(L^2 + \hbar^2) \tag{5}$$

$$\mathbf{L} \cdot \mathbf{A} + \mathbf{A} \cdot \mathbf{L} = 0. \tag{6}$$

The last two expressions are analogous to the classical results. To simplify the notation, let's work with natural units where $\hbar = m_e = a_0 = 1$. Here, m_e is the mass of the electron and

a_0 is the Bohr radius. In these units, the energy and momentum scales are

$$[\text{energy}] = \frac{\hbar^2}{ma^2}, \quad [\text{momentum}] = \frac{\hbar}{a}$$

respectively, and from the definition of the Bohr radius,

$$a = \frac{\hbar^2}{m_e \gamma} \Rightarrow \gamma = 1$$

In natural units,

$$H = \frac{1}{2}\mathbf{p}^2 - \frac{1}{r}$$

and

$$\mathbf{A} = \frac{1}{2}\left(\mathbf{p} \times \mathbf{L} - \mathbf{L} \times \mathbf{p}\right) - \frac{\mathbf{r}}{r}$$

Restricting to eigenstates of H, i.e., $H|E\rangle = E|E\rangle$,

$$A^2 = 1 + 2E(L^2 + 1)$$

or

$$-\frac{A^2}{2E} = -\frac{1}{2E} - L^2 - 1$$

For bound states $E < 0$ define the scaled Runge–Lenz vector as

$$\mathbf{K} \equiv \frac{\mathbf{A}}{\sqrt{-2E}}.$$

It follows that

$$K^2 + L^2 = -\frac{1}{2E} - 1$$

where

$$[L_i, K_j] = i\epsilon_{ijk}K_k, \quad \text{and} \quad [K_i, K_j] = i\epsilon_{ijk}L_k.$$

Now define operators $\mathbf{M}$ and $\mathbf{N}$ that "factorize" the operator $K^2 + L^2$:

$$\mathbf{M} \equiv \frac{1}{2}(\mathbf{L} + \mathbf{K}), \quad \text{and} \quad \mathbf{N} \equiv \frac{1}{2}(\mathbf{L} - \mathbf{K}).$$

The operators $\mathbf{M}$ and $\mathbf{N}$ both commute with the Hamiltonian, and furthermore, they have the same commutation relations as two independent angular momentum operators:

$$[M_i, M_j] = i\epsilon_{ijk}M_k, \quad \text{and} \quad [N_i, N_j] = i\epsilon_{ijk}N_k$$

where the final relation $[M_i, N_j] = 0$ establishes their independence. In terms of M^2 (or equivalently in terms of N^2), we find

$$M^2 = N^2 = \frac{1}{4}\left(K^2 + L^2\right) = -\frac{1}{4}\left(\frac{1}{2E} + 1\right)$$

In the shared eigenspace of H and M^2 (or N^2), there exists $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$ such that

$$j(j+1) = -\frac{1}{4}\left(\frac{1}{2E} + 1\right)$$

$$-\frac{1}{2E} = 4j^2 + 4j + 1 = (2j+1)^2$$

Letting $2j + 1 = n$ for some integer n, we finally obtain

$$E = -\frac{1}{2n^2}$$

Restoring physical units, we finally arrive at the energy levels of the hydrogen atom:

$$E_n = -\frac{13.6\,\mathrm{eV}}{n^2}.$$

In principle, we could go on to generate the eigenstates using the ladder operators $M_\pm$ and $N_\pm$, but in order to write them in terms of the familiar eigenstates $|n\ell m\rangle$, we need to know how to couple (add) angular momenta. The addition of angular momentum will be introduced in chapter #10 and discussed more comprehensively in Phys 438B.